MEGHANA B

B.E. - Artificial Intelligence & Machine Learning

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PROFESSIONAL SUMMARY

An enthusiastic, self-motivated and detail-oriented computer science pursuing student who would like to learn new technologies and concepts in the practical way. Also, passionate person who wish to upgrade skills consistently that industry demands in this rapidly changing innovative world. An eager to apply my technical skills and creativity in developing high quality software and contribute to the success of the organization

KEY EXPERTISE

Java | React.js | Neo4j | Python | Web Development | Software Engineering | DSA | DBMS | Git | FastAPI | MySQL | Problem Solving | OOP | Operating Systems | Computer Networking | Machine Learning | Artificial Intelligence | Node.js | HTML | CSS | Javascript | Github | Django | RESTful API

EDUCATION

RNS Institute of Technology 2023 - 2027

B.E. - Artificial Intelligence & Machine Learning | CGPA: **8.94** / **10**

RK vision pu college, chikkaballapur 2023

12th | PUE Karnataka | Percentage: **93.50** / **100**

Sri Bhairaveshwara Vidya Samaste, Chikkaballapur 2021

10th | KSEEB | Percentage: **96** / **100**

PROJECTS

LLM-Based Research Paper Knowledge Graph Explorer 23 Mar, 2026 - 09 Jul, 2026

Team Size: 2

Key Skills: Neo4j | Python | LLM | FastAPI | PageRank Algorithm | React | TypeScript | Sigma.js

Project Link: https://github.com/Meghana-B266/research-kg-explorer.git

Designed and built a full-stack AI knowledge-graph platform that extracts structured entities from research PDFs using a locally-hosted LLM (Ollama), eliminating cloud API costs and keeping all data on-device.
Implemented a graph-based data model in Neo4j using MERGE-based Cypher queries to deduplicate authors, keywords, and methods across hundreds of relationships, enabling queries a relational database couldn't handle efficiently.
Built automated claim-verification, negative-result-detection, and cross-document hypothesis-generation features, using LLM-driven background analysis to surface research insights a manual read would likely miss.
Diagnosed and resolved full-stack production defects — including a Neo4j version-incompatible Cypher syntax error and a missing dependency chain — through systematic root-cause analysis of server stack traces.

Deep Learning Approach for Detecting Floating Plastics in Riverine Systems 14 Oct, 2025 - 10 Dec, 2025

Mentor: prof subhashini baglkot | **Team Size:** 3

Key Skills:

Python | debugging and code integration | SQLite | Problem Solving | Java | Artificial Intelligence | Deep Learning | FastAPI | YOLOv11n | OpenCV | HTML | CSS | Javascript

Project Link: https://github.com/Meghana-B266/waste_detection.git

Built a real-time river pollution monitoring system using YOLOv11n for floating plastic detection
• Implemented Byte Track multi-object tracking to improve detection stability and reduce false positives
• Integrated IP camera video streaming and real-time inference at 15 FPS on CPU
• Automated email alert system using SMTP with annotated detection images
• Logged detection events using SQLite for historical analysis and reporting
• Achieved 98.8% mAP@50, 95% F1-score on a custom dataset of 2,200+ plastic instances

PUBLICATIONS / RESEARCH / WHITE PAPERS

Deep Learning-Based Detection of Surface Plastic Pollution in Riverine Systems

24 Jul, 2026

Conference Paper | International Conference for Women in Innovation, Technology and Entrepreneurship (ICWITE 2026) |

Mentor: Suhasini Bagalkot | **No. of Authors:** 4

Key Skills: Deep Learning | yolov11 | computer vision | ByteTrack | python | object Detection | Machine Learning

Developed a deep learning-based real-time system for detecting and tracking floating plastic pollution in riverine environments using YOLOv11n and ByteTrack. The system processes camera feeds for continuous monitoring and automatically sends email alerts upon confirmed plastic detection, achieving 98.8% detection performance.

ASSESSMENTS / CERTIFICATIONS

Design and analysis of Algorithms

Provider: Coursera

Aggregate: 92.25 / 100

Subjects: Analysis and design of algorithm (92.25 / 100)

Key Skills:

Data Structures | Searching Algorithms | Sorting Algorithms | Indexing | Algorithmic Problem Solving | Computational Thinking | Algorithm Efficiency

Completed a course focused on fundamental algorithms and techniques for searching, sorting, and indexing, strengthening problem-solving and algorithmic thinking skills for efficient data processing.

Database management systems

Provider: Coursera

Aggregate: 84.12 / 100

Subjects: DBMS (84.12 / 100)

Key Skills:

SQL | Database Management | Relational Databases | Data Analysis | Python | SQL Queries | Data Retrieval | Data Manipulation | Database Concepts

Completed an IBM-authorized course covering database concepts, SQL, and Python-based database interaction for data science applications, with emphasis on querying and analyzing data using SQL and Python.

Introduction to Artifical Intelligence

Provider: Coursera

Aggregate: 98 / 100

Subjects: AI (98 / 100)

Key Skills: Artificial Intelligence | AI Fundamentals | Machine Learning Fundamentals | AI Applications | Problem Solving

Completed an introductory course in Artificial Intelligence, gaining foundational knowledge of AI concepts, techniques, and applications.

Java Full Stack Developer

Provider: Coursera

Aggregate: 92 / 100

Subjects: java (92 / 100)

Key Skills:

Java | Front-End Web Development | Object-Oriented Programming (OOP) | Data Structures | HTML | CSS | Spring | Spring Boot | Backend Development | Web Application Development

Completed a 3-course Java Full Stack Development specialization covering Java programming, frontend web development using HTML, CSS and Angular, and backend development using Spring and Spring Boot. Gained foundational knowledge of data structures and skills required to develop Java-based web applications.

Machine Learning with Python

Provider: IBM Skillsbuild

Aggregate: 92 / 100

Subjects: Machine learning (92 / 100)

Key Skills:

Machine Learning | Supervised & Unsupervised Learning | Regression & Classification | Data Preprocessing & EDA | Feature Engineering | Decision Trees & Random Forest | KNN, SVM & Naive Bayes | Model Evaluation | Cross-Validation | Hyperparameter Tuning | Ensemble Learning | PCA & Dimensionality Reduction

Completed IBM's Machine Learning with Python course covering fundamental machine learning concepts, including supervised and unsupervised learning, data preprocessing, model development, evaluation, and implementation of ML algorithms using Python and scikit-learn.

Introduction to Generative AI

Provider: Google Cloud / Coursera
Aggregate: 85 / 100
Subjects: Generative AI (85 / 100)
Key Skills:

Generative AI | Artificial Intelligence | AI Fundamentals | Generative AI Applications | Large Language Models (LLMs) | AI Concepts

Completed an introductory course in Generative AI, developing foundational knowledge of generative AI concepts, applications, and technologies.

SEMINARS / TRAININGS / WORKSHOPS

Deloitte Technology Job Simulation

Institute Name: Deloitte
03 Feb, 2026 - 03 Feb, 2026

Key Skills: Coding | Software Development

Completed a practical technology job simulation focused on coding and development, gaining hands-on exposure to real-world technology tasks.

Deloitte Data Analytics Job Simulation

03 Feb, 2026 - 03 Feb, 2026
Institute Name: Deloitte

Key Skills: Data Analysis | Forensic Technology

Completed a practical data analytics job simulation involving data analysis and forensic technology tasks.

WEB LINKS

- Github - https://github.com/Meghana-B266

PERSONAL DETAILS

Gender: Female
Marital Status: Single
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Date of Birth: 02 Jun, 2006
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